Fidelity total anticoherence is nondecreasing under covariant noise at order two

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Abstract

We prove the fidelity part of Conjecture 1 of *Total, quantum, and classical measures of anticoherence for mixed spin states*, at $t = 2$, for every $N \geq 3$. The notation and the squared Uhlmann–Jozsa fidelity normalization are those of `resub_v0.tex`. The proof is based on `su2_t2_full_direct_proof.pdf`. It treats arbitrary covariant channels directly, derives their two relevant multipliers without $6j$ symbols, and ends with an elementary three-eigenvalue argument.

1 Statement and conventions

As in the manuscript, $\mathcal{H}_j$ is the spin-j Hilbert space, $N = 2j$, and ρ_S is the state corresponding to $\rho \in \mathcal{D}(\mathcal{H}_j)$ under the identification $\mathcal{H}_j \simeq \mathrm{Sym}^N(\mathbb{C}^2)$. Put

$$\rho_t = \mathrm{Tr}_{N-t}(\rho_S), \qquad \rho_0^{(t)} = \frac{\mathbb{1}_{t+1}}{t+1}, \qquad F(\rho_t, \rho_0^{(t)}) = \frac{(\mathrm{Tr}\,\sqrt{\rho_t})^2}{t+1}.$$

Thus the fidelity total candidate of the manuscript is

$$\mathcal{A}_t^{T,F}(\rho) = \frac{(\mathrm{Tr}\,\sqrt{\rho_t})^2 - 1}{t}. \tag{1}$$

All determinants and traces of two-qubit reductions below are taken on the three-dimensional symmetric subspace, including any zero eigenvalues.

Theorem 1 (Conjecture 1, fidelity candidate, $t = 2$)**.** *Let $N \geq 3$, $j = N/2$, and let Φ be a completely positive, trace-preserving map on $\mathcal{B}(\mathcal{H}_j)$ satisfying*

$$\Phi(U\rho U^\dagger) = U\Phi(\rho)U^\dagger \quad \textit{for every spin-}j\textit{ rotation } U \textit{ and every state } \rho.$$

Then, for every $\rho \in \mathcal{D}(\mathcal{H}_j)$,

$$\boxed{\mathcal{A}_2^{T,F}(\Phi(\rho)) \geq \mathcal{A}_2^{T,F}(\rho).} \tag{2}$$

The condition $N \geq 3$ is exactly the proper-order condition $2 \leq t < N$ at $t = 2$. No restriction to pure inputs or to random rotations is made. The theorem does not address the trace-distance candidate.

2 The two channel multipliers

Use the Hilbert–Schmidt orthonormal tensor operators T_{LM} of the manuscript and put

$$f = f_1, \qquad g = f_2, \qquad S = N(N+2), \qquad c = j(j+1) = S/4.$$

We give an elementary justification that these two real multipliers exist; no decomposition of the Choi representation is needed. Let $H = J_z$, $E = J_x + iJ_y$, $E^\dagger = J_x - iJ_y$. In the usual weight basis $|k\rangle$, $0 \le k \le N$,

$$H\,|k\rangle = (k-j)\,|k\rangle\,, \qquad E\,|k\rangle = \sqrt{(k+1)(N-k)}\,|k+1\rangle\,.$$

State covariance extends to all operators by complex linearity: positive operators are non-negative multiples of states, and every matrix is a complex linear combination of positive operators. Differentiating rotation covariance shows that Φ commutes with $[J_a, \cdot]$, hence with $[H, \cdot]$, $[E, \cdot]$, and $[E^\dagger, \cdot]$. If $[H, X] = LX$ and $[E, X] = 0$, then $X\,|0\rangle$ is a multiple of $|L\rangle$. Moreover $XE^k\,|0\rangle = E^k X\,|0\rangle$ determines X on the entire basis. Thus X is a multiple of E^L for $L = 0, 1, 2$. Applying this to $X = \Phi(E^L)$ shows that Φ is scalar on each space generated from E^L by repeated commutators with $E^\dagger$. For $L = 0$ this is the scalar space; for $L = 1$ it is the span of J_a; for $L = 2$ it is the span of

$$Q_{ab} = \frac{J_a J_b + J_b J_a}{2} - \frac{c}{3}\delta_{ab}\mathbb{1}.$$

For example,

$$[E^\dagger, E] = -2H, \quad [E^\dagger, E^2] = -2(HE + EH), \quad [E^\dagger, [E^\dagger, E^2]] = 12H^2 - 4c\mathbb{1};$$

these identities, their adjoints, and $\sum_a Q_{aa} = 0$ give all the listed components. Hermiticity preservation makes the scalars real, and trace preservation makes the rank-zero scalar one. Hence

$$\Phi(\mathbb{1}) = \mathbb{1}, \qquad \Phi(J_a) = fJ_a, \qquad \Phi(Q_{ab}) = gQ_{ab}.$$

The trace adjoint also commutes with these commutators and preserves Hermiticity. The same argument applies to it, and trace pairing with any nonzero Hermitian component shows that its rank-one and rank-two scalars are the same f, g. This also justifies the transformation of expectation values used below.

Lemma 1 (Multiplier bounds)**.** *Every channel in Theorem 1 satisfies*

$$-\frac{2}{3} \le g \le 1, \qquad 3|f| \le 2 + g, \tag{3}$$

$$3Sf^2 - 6f - S \le 2(S-3)g. \tag{4}$$

Consequently $|f| \le 1$ *and* $|g| \le 1$*; if* $g < -1/2$*, then* $|f| < 1/3$*.*

Proof. Set $d = N + 1$ and let $C_\Phi \succeq 0$ be the unnormalized Choi matrix, with convention

$$\mathrm{Tr}[C_\Phi(B \otimes C^{\mathsf{T}})] = \mathrm{Tr}[B\Phi(C)]. \tag{5}$$

Complete positivity gives $C_\Phi \succeq 0$ by applying $\Phi \otimes \mathrm{id}$ to the unnormalized maximally entangled projector. Trace preservation gives $\mathrm{Tr}\, C_\Phi = d$. Thus $\omega = C_\Phi/d$ is a density matrix. Define the Hermitian observable

$$X = -\sum_{a=1}^{3} J_a \otimes J_a^{\mathsf{T}}.$$

Only the following two polynomial identities are required:

$$\sum_{a,b} Q_{ab} \otimes Q_{ab}^{\mathsf{T}} = X^2 + \frac{X}{2} - \frac{c^2}{3}\mathbb{1}, \qquad \sum_{a,b} Q_{ab}^2 = \frac{c(4c-3)}{6}\mathbb{1}. \tag{6}$$

For the first identity, put $B_{ab} = (J_aJ_b + J_bJ_a)/2$ and expand:

$$\sum_{a,b} B_{ab} \otimes B_{ab}^{\mathsf{T}} = X^2 + \frac{1}{2}\sum_{a,b} J_aJ_b \otimes [J_a, J_b]^{\mathsf{T}} = X^2 + \frac{X}{2}.$$

The last equality uses $\sum_{a,b} \epsilon_{abk} J_aJ_b = iJ_k$; subtracting the scalar trace parts gives the first identity. For the second, expansion of $\sum_b [J_b, [J_b, J_a]] = 2J_a$ gives $\sum_b J_bJ_aJ_b = (c-1)J_a$. Thus, for $B_{ab} = (J_aJ_b + J_bJ_a)/2$,

$$\sum_{a,b} B_{ab}^2 = \frac{1}{2}\sum_{a,b}(J_aJ_bJ_aJ_b + J_aJ_bJ_bJ_a) = \left(c^2 - \frac{c}{2}\right)\mathbb{1},$$

which proves the second identity. The trace pairing and the two multipliers now give

$$\mathrm{Tr}(\omega X) = -cf, \qquad \mathrm{Tr}(\omega X^2) = \frac{c(4c-3)}{6}g + \frac{cf}{2} + \frac{c^2}{3}.$$

Nonnegative variance, $\mathrm{Tr}(\omega X^2) \geq [\mathrm{Tr}(\omega X)]^2$, is precisely (4) after multiplication by $12/c$.

The affine marginal law (10), established in the next section, applied to a spin-coherent input gives reduced output eigenvalues

$$\frac{2+g+3f}{6}, \qquad \frac{2+g-3f}{6}, \qquad \frac{1-g}{3}.$$

Their nonnegativity gives $g \leq 1$ and $3|f| \leq 2+g$. Finally, if $g < -2/3$, (4) and $S \geq 15$ would give

$$0 > 9(Sf-1)^2 + S^2 - 12S - 9 = 9(Sf-1)^2 + (S-15)(S+3) + 36 \geq 0.$$

This proves (3). If $g < -1/2$, (4) implies $Sf^2 - 2f - 1 < 0$. Since $S \geq 15$, this forces $-1/3 < f < 1/3$: otherwise $Sf^2 - 2f - 1 \geq 15f^2 - 2f - 1 = (3f-1)(5f+1) \geq 0$. □

3 Physical two-qubit reductions

Write the symmetric two-qubit reduction in its Pauli form

$$\rho_2 = \frac{1}{4}\left[\mathbb{1}\otimes\mathbb{1} + \sum_{a=1}^{3} r_a(\sigma_a\otimes\mathbb{1} + \mathbb{1}\otimes\sigma_a) + \sum_{a,b=1}^{3} T_{ab}\sigma_a\otimes\sigma_b\right]. \tag{7}$$

Here $\mathbf{r} \in \mathbb{R}^3$, $T = T^{\mathsf{T}} \in M_3(\mathbb{R})$, and $\mathrm{Tr}\, T = 1$, since $\sum_a \sigma_a\otimes\sigma_a = 2\mathsf{F} - \mathbb{1}$ and the swap F is the identity on the symmetric subspace. In the orthonormal Cartesian spin-one basis

$$|x\rangle = \frac{|11\rangle - |00\rangle}{\sqrt{2}}, \quad |y\rangle = \frac{i(|11\rangle + |00\rangle)}{\sqrt{2}}, \quad |z\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

put $A = \mathbb{1}_3 - T$ and let $[\mathbf{r}]_\times\mathbf{v} = \mathbf{r}\times\mathbf{v}$. Substitution in (7) gives

$$2\rho_2 = A + i[\mathbf{r}]_\times, \qquad 8\det\rho_2 = \det A - \mathbf{r}^{\mathsf{T}}A\mathbf{r}. \tag{8}$$

To check the determinant identity, diagonalize A by a proper real rotation. The matrix then has diagonal a, b, c and off-diagonal entries $-ir_3, ir_2, -ir_1$ and their conjugates; its determinant is $abc - ar_1^2 - br_2^2 - cr_3^2$.

There is a useful bound specific to reductions of symmetric N-qubit states. With $J_a = \frac{1}{2}\sum_{n=1}^{N}\sigma_a^{(n)}$, permutation symmetry gives

$$\langle J_a\rangle = \frac{N}{2}r_a, \qquad \left\langle\frac{J_aJ_b + J_bJ_a}{2}\right\rangle = \frac{N}{4}\delta_{ab} + \frac{N(N-1)}{4}T_{ab}.$$

The state has a symmetric three-qubit reduction with the same two-qubit marginal. Applying these identities to that three-qubit state and using nonnegative variance of its collective spin gives $\mathbb{1} + 2T - 3\mathbf{r}\mathbf{r}^{\mathsf{T}} \succeq 0$. Hence $\mathbb{1} + 2T \succeq 0$. Testing (8) on real vectors also gives $A \succeq 0$. Therefore

$$0 \preceq A \preceq \frac{3}{2}\mathbb{1}_3, \qquad \operatorname{Tr} A = 2. \tag{9}$$

This uses only the symmetric three-qubit extension, available for every $N \geq 3$.

Write $\rho' = \Phi(\rho)$. The partial trace intertwines rotations, so the rank-one and rank-two parts of the reduction acquire the same multipliers $f_1 = f$ and $f_2 = g$ as the global tensor components. Equivalently, the above first and second moments give directly

$$\mathbf{r}' = f\mathbf{r}, \quad T' = gT + \frac{1-g}{3}\mathbb{1}_3, \quad A' = \alpha\mathbb{1}_3 + gA, \qquad \alpha = \frac{2(1-g)}{3}. \tag{10}$$

4 Determinant monotonicity

We prove two elementary estimates for A and A'.

Lemma 2. *Under (9) and (3),* $\det A' \geq \det A$.

Proof. Write the eigenvalues of A as $2/3 + x_i$, where $\sum_i x_i = 0$, and set $s_2 = \sum_i x_i^2$, $p = x_1x_2x_3$. Expansion gives

$$\det A' - \det A = \frac{1-g^2}{3}s_2 - (1-g^3)p. \tag{11}$$

We claim $p \leq 5s_2/36$. This is immediate if $p \leq 0$. If $p > 0$, exactly one x_i is positive, so after relabeling $(x_1, x_2, x_3) = (q + r, -q, -r)$ with $q, r > 0$. The upper eigenvalue bound gives $q + r \leq 5/6$, whereas $s_2 = 2(q^2 + qr + r^2) \geq 6qr$. Hence $p = (q + r)qr \leq 5s_2/36$, as claimed. Since $g \leq 1$, (11) now gives

$$\det A' - \det A \geq \frac{(1-g)s_2}{36}(7 + 7g - 5g^2) \geq 0,$$

because $7 + 7g - 5g^2 = 1/9 + (g + 2/3)(31/3 - 5g) \geq 1/9$ on $[-2/3, 1]$. □

Lemma 3 (Cofactor comparison)**.** *If b, c are any two eigenvalues of A, and $b' = \alpha + gb$, $c' = \alpha + gc$, then*

$$b'c' \geq f^2 bc. \tag{12}$$

Proof. We have $0 \leq b, c \leq 3/2$ and $b + c \leq 2$. Also $\alpha + gx \geq 0$ for $0 \leq x \leq 3/2$: for $g \geq 0$ this is immediate, and for $g < 0$ it follows from

$$\alpha + gx \geq \alpha + \frac{3}{2}g = \frac{4 + 5g}{6} \geq \frac{1}{9}. \tag{13}$$

Write $s = b + c$, $p = bc$.

If $g \geq -1/2$, then $2g + \alpha = 2(1 + 2g)/3 \geq 0$. The identity

$$4\big[b'c' - (g + \alpha)^2 bc\big] = \alpha\big[(2g + \alpha)(b - c)^2 + (2 - s)\{2\alpha + (2g + \alpha)s\}\big] \geq 0$$

and $(g + \alpha)^2 = (2 + g)^2/9 \geq f^2$ prove the claim.

If $-2/3 \leq g < -1/2$, Lemma 1 gives $f^2 < 1/9$. Set $h = g^2 - 1/9 > 0$. The inequality $(3/2 - b)(3/2 - c) \geq 0$ gives $p \geq 3s/2 - 9/4$. Furthermore

$$\alpha g + \frac{3}{2}h = \frac{(5g - 1)(g + 1)}{6} < 0.$$

It follows that

$$\begin{aligned} b'c' - \frac{bc}{9} &= \alpha^2 + \alpha g s + hp \\ &\geq \alpha^2 - \frac{9}{4}h + \left(\alpha g + \frac{3}{2}h\right) s \\ &\geq \alpha^2 + 2\alpha g + \frac{3}{4}h = \frac{13 + 16g - 5g^2}{36} \geq \frac{1}{324} > 0. \end{aligned}$$

The last bound follows from $13 + 16g - 5g^2 = 1/9 + (g + 2/3)(58/3 - 5g)$. This proves (12) in both cases. □

We can now show

$$\det \rho_2' \geq \det \rho_2. \tag{14}$$

In a common eigenbasis of A, A', with respective eigenvalues a, b, c and a', b', c', all six eigenvalues are nonnegative. Set $\Delta = 8 \det \rho_2$, $\Delta' = 8 \det \rho_2'$. Equation (8) gives the polynomial identity

$$\begin{aligned} abc(\Delta' - \Delta) = (a'b'c' - abc)\Delta \\ + aa' r_1^2 (b'c' - f^2 bc) + bb' r_2^2 (a'c' - f^2 ac) + cc' r_3^2 (a'b' - f^2 ab) \geq 0. \end{aligned}$$

Every term is nonnegative by Lemmas 2 and 3, and $\Delta \geq 0$ follows from $\rho_2 \succeq 0$. If $abc > 0$, division proves (14). If A is singular, then (8), $A \succeq 0$ and $\det \rho_2 \geq 0$ force $\det \rho_2 = 0$. The inequality is then immediate from $\rho_2' \succeq 0$. This also covers all boundary states.

5 From determinant and purity to fidelity

Orthogonally decompose the reduction into its tensor ranks:

$$\rho_2 = \frac{\mathbb{1}_3}{3} + X_1 + X_2, \qquad \rho_2' = \frac{\mathbb{1}_3}{3} + f X_1 + g X_2.$$

The Hermitian X_1, X_2 are traceless and Hilbert–Schmidt orthogonal. The bounds $|f|, |g| \leq 1$ therefore imply

$$\operatorname{Tr}[(\rho_2')^2] = \frac{1}{3} + f^2 \operatorname{Tr}(X_1^2) + g^2 \operatorname{Tr}(X_2^2) \leq \operatorname{Tr}(\rho_2^2). \tag{15}$$

For a positive trace-one 3×3 matrix B, let $e_2(B) = (1 - \operatorname{Tr} B^2)/2$ and $e_3(B) = \det B$. Equations (14) and (15) show that both e_2 and e_3 increase. The following elementary fact completes the argument without an integral representation of the square root.

Lemma 4 (Three-eigenvalue comparison)**.** *If $B, C \succeq 0$ are 3×3 matrices of trace one with $e_2(C) \geq e_2(B)$ and $\det C \geq \det B$, then $\operatorname{Tr} \sqrt{C} \geq \operatorname{Tr} \sqrt{B}$.*

Proof. Let $x, y, z \geq 0$ be the square roots of the eigenvalues of B. Write $v = x + y + z$, $d = xyz = \sqrt{\det B}$ and $e = e_2(B)$. Since $x^2 + y^2 + z^2 = 1$, elementary expansion gives

$$v \geq 1, \qquad (v^2 - 1)^2 = 4e + 8dv, \qquad v(v^2 - 1) \geq 18d. \tag{16}$$

For the last inequality, use $v^2 - 1 = 2(xy + xz + yz)$ and

$$v(xy + xz + yz) - 9xyz = z(x - y)^2 + x(y - z)^2 + y(z - x)^2 \geq 0.$$

Define u, d', e' in the same way for C. If $u < v$, then $u \geq 1$, $d' \geq d$ and $e' \geq e$ imply

$$(v^2 - 1)^2 - (u^2 - 1)^2 = 4(e - e') + 8(dv - d'u) \leq 8d(v - u).$$

Dividing by $v-u>0$ gives $(v+u)(v^2+u^2-2)\le 8d$. But

$$(v+u)(v^2+u^2-2)=v(v^2-1)+u(v^2-1)+(v+u)(u^2-1)>v(v^2-1)\ge 18d\ge 8d,$$

a contradiction. Here strictness follows from $u\ge 1$ and $v>u$. □

Apply Lemma 4 to $B=\rho_2$, $C=\rho_2'$. Then $\operatorname{Tr}\sqrt{\rho_2'}\ge\operatorname{Tr}\sqrt{\rho_2}\ge 0$; squaring and using (1) proves (2). This proves precisely the fidelity part of Conjecture 1 at $t=2$. □

6 Formal verification and reproducibility

The accompanying Lean 4.24.0 development proves Theorem 1 from its stated physical hypotheses. The public result is `SU2T2.conjecture1_fidelity_t2` in `lean/FullTheorem.lean`. Its only assumptions are $N\ge 3$, complete positivity and trace preservation of a complex-linear map, rotation covariance on states, and positivity and unit trace of the input.

The definitions use the normalized Dicke embedding into the actual N-qubit tensor product, matrices in SU(2), genuine partial traces, and the positive matrix square root from continuous functional calculus. The formal proof derives the infinitesimal covariance relations, the Choi variance bound, the physical marginal constraints and affine law, and the spectral identification. None of these steps remains an additional hypothesis. In particular, the proof requires neither a Clebsch–Gordan decomposition nor an assumed scalar reduction.

The complete theorem and all its bridges depend only on the standard Lean/mathlib foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no unfinished proofs or additional axioms. Dependency pins, build commands, the axiom audit, and checked-source hashes are supplied in `lean/README.md`, `lean/verification.log`, `lean/axiom-audit.log`, and `lean/source-sha256.txt`. The original scalar certificate remains available as a subsidiary result. The file `SpinAction.lean` also proves invariance of the Dicke subspace under every tensor-power rotation, the spin-action group law, and unitarity. The resulting `unitaryAction` is a group homomorphism. Lean uses ascending magnetic order $m=k-N/2$; the manuscript's numerical arrays use descending order $m=N/2-q$. The conversion is $q=N-k$: $v_k^{\mathrm{asc}}=v_{N-k}^{\mathrm{desc}}$ and $\rho_{kl}^{\mathrm{asc}}=\rho_{N-k,N-l}^{\mathrm{desc}}$. For the reversal matrix R, channels convert as $\Phi^{\mathrm{asc}}(A)=R\Phi^{\mathrm{desc}}(RAR)R$. The vector and matrix reversals are proved involutive in `BasisOrder.lean`; see `BASIS_ORDER.md` for details.

Remark 1 (Why the marginal constraint matters)**.** *The bound $A\preceq 3\mathbb{1}_3/2$ comes from the symmetric extension to $N\ge 3$ qubits. An arbitrary spin-one density matrix need not obey it. The proof therefore does not assume that the induced reduced map is a positive or completely positive map on the whole spin-one state space.*